

A STUDENT-FRIENDLY NLP PROJECT

UniLex

Complex terms, explained like a classmate would.

COMPUTER SCIENCE

ARTIFICIAL
INTELLIGENCE

A web-based academic dictionary powered by NLP search.

Live demo: unilex-samar.onrender.com



The problem

Technical terms can slow learning down.

Students meet unfamiliar CS and AI vocabulary in lectures, assignments, and textbooks. Formal definitions can be difficult to understand and often lack simple examples.

WHAT STUDENTS NEED

- 01 **Simple explanations**
- 02 **Practical examples**
- 03 **Related concepts**
- 04 **Easy search by meaning**

Meet UniLex

One place to look up a concept, understand it, and connect it to other ideas.

01

Keyword search

Find terms by matching words.

02

Smart search

Find related concepts by meaning.

03

Lecture extraction

Paste notes to identify known terms.

Each entry can include a student-friendly explanation, formal definition, example, and related terms.

How it works

01

Prepare the glossary

Collect, clean, and standardize CS and AI terms.

02

Search with NLP

TF-IDF + cosine similarity for keyword search; sentence embeddings for semantic search.

03

Present useful context

Show explanations, definitions, examples, and related terms.

Built with Python, HTML, CSS, and JavaScript; deployed online through Render.

A simple user journey

Search a term

Example: “Overfitting” returns a simplified explanation and formal definition.

OVERFITTING

Learns training data too closely; performs poorly on unseen data.

Explore lecture notes

Paste a paragraph and UniLex identifies terms already available in its glossary.

Machine
Learning

Overfitting

Testing
concepts

The goal: make technical learning clearer and more independent.

Value today — and what comes next

WHAT UNILEX OFFERS

- Accessible CS & AI glossary
- Multiple search approaches
- Explanations and examples
- Lecture-text term extraction

NEXT IMPROVEMENTS

- Expand and review the glossary
- Evaluate relevance with more queries
- Collect and analyze student feedback
- Improve handling of unfamiliar wording

We will report measured outcomes only after systematic testing and survey analysis.

THANK YOU

Make complex terms feel simple.

Explore UniLex

<https://unilex-samar.onrender.com/>

Source code: github.com/sara78606/UniLex



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